# Probabilistic neural surrogates for uncertainty-to-risk analysis in coupled multi-physics systems: application to a heat-pipe-cooled reactor

**用于耦合多物理系统不确定性到风险分析的概率神经代理模型：以热管冷却反应堆为例**

## 1. Introduction

Complex systems are commonly shaped by coupled mechanisms, incomplete knowledge and expensive computation. In such systems, uncertainty is a prerequisite for credible assessments and decision support. This point this particularly sharp in safety related analysis, where uncertainty affects not only the spread pf response range, but also the interpretation of safety margins and the admissible region of design and operating parameters. The central issue is therefore not merely how to predict a coupled response, but how to convert uncertain inputs into risk-relevant conclusions under explicit safety criteria. This uncertainty-to-decision framing is used to support inference, model selection and downstream decisions, and possesses broad generalizability.

As a typical example of such complexity, Heat Pipe Cooled Reactors (HPRs) represent a transformative class of Small Modular Reactors (SMRs), offering high-density power for remote and space applications through passive heat removal. However, this intentional compactness gives rise to significant safety risks associated with strong multi-physics coupling—specifically, the intricate interconnections between neutronic reactivity, thermal conduction, and mechanical deformation create a dynamic feedback system where perturbations in one field can propagate and exacerbate risks in the others.

作为此类复杂性的典型代表，热管冷却反应堆（HPRs）是一类具有变革性的小型模块化反应堆（SMRs），通过被动散热为偏远及空间应用提供高密度功率。然而，这种刻意设计的紧凑性会引发与强多物理场耦合相关的显著安全风险——具体而言，中子反应性、热传导和机械变形之间的复杂关联形成了一个动态反馈系统，其中一个场的扰动会传播并加剧其他场的风险。

Previous studies have shown that thermal stress is a well-recognized critical safety concern, with a high risk of exceeding design limits. Within the monolith-type core, small epistemic uncertainties in material properties are amplified through complex multi-physics feedback loops, thereby posing substantial threats to structural integrity. It is worth emphasizing that many material parameters used

in the thermo-mechanical calculations of the matrix are derived from empirical data; however, the uncertainty ranges of these empirical parameters and their potential impacts on the accuracy of thermo-mechanical analysis remain unclear and unquantified. This lack of clarity regarding the uncertainty of key material parameters further exacerbates the ambiguity in assessing structural safety risks of the monolith core, making uncertainty analysis an indispensable and critical step for the safe and reliable operation of HPR systems. Nevertheless, a systematic framework for mapping uncertainty to risk remains underdeveloped, leaving a critical gap in converting fundamental material fluctuations into quantifiable reactor-level safety margins.

以往研究表明，热应力通常是HPR运行中的主要安全瓶颈。在整体式堆芯内，材料特性中的微小认知不确定性会通过复杂的多物理场反馈回路被放大，进而对结构完整性构成重大威胁。需要强调的是，基体热力计算中使用的许多材料参数均来源于经验数据，但这些经验参数的不确定性范围及其对热力分析准确性的潜在影响仍不明确且未被量化。这种关键材料参数不确定性的不明确性，进一步加剧了整体式堆芯结构安全风险评估的模糊性，使得不确定性分析成为热管冷却反应堆（HPR）系统安全可靠运行不可或缺的关键环节。然而，用于将不确定性映射到风险的系统框架仍不完善，这使得在将基本材料波动转化为可量化的反应堆级安全裕度方面存在重大缺口。

High-fidelity multi-physics simulations are essential for resolving these complex multi-field interactions, but they are inherently unsuitable for repeated-computation workflows. These workflows, including uncertainty propagation, sensitivity analysis, and parameter exploration, require a large number of iterative simulations that exceed the practical computational capacity of deterministic high-fidelity solvers.

高保真多物理场模拟对于解决这些复杂的多场相互作用至关重要，但它们本质上不适合多次计算工作流。这些工作流（包括不确定性传播、敏感性分析和参数探索）需要大量的迭代模拟，而这超出了确定性高保真求解器的实际计算能力。

Conventional neural network-based surrogate models often prioritize pointwise accuracy over the sufficient integration of physical information required for safety-critical inference. Moreover, Physics-Informed Neural Networks (PINNs), which have gained increasing popularity recently, have high requirements for hyperparameter tuning and model training, making them prone to training instability and non-convergence issues. To address this computational dilemma, this study uses a neural framework incorporated with physical regularization terms, aiming to bridge the gap between high-fidelity physics modeling and rapid, risk-informed decision support.

传统基于神经网络的替代模型往往优先考虑逐点精度，而忽视了安全关键推理所需的、对物理信息的充分融合。此外，近年来备受关注的物理信息神经网络（PINNs），对超参数调优和模型训练要求较高，易出现训练不稳定、难以收敛的问题。为解决这一计算困境，本研究使用了一种加入物理正则项的神经框架，旨在弥合高保真物理建模与快速、风险导向决策支持之间的差距。
【还要补研究现状，主要是热管堆的，多物理耦合的，不确定性分析的，三大类，可以参考我的老文章 rpha】

Here we establish an end-to-end workflow for uncertainty-to-risk analysis in a coupled HPR system. The workflow uses a probabilistic neural surrogate to emulate a high-fidelity neutron–thermal–mechanical mapping. We then use the surrogate for three connected tasks: forward uncertainty propagation under a fixed input domain, stress-risk analysis and observation-driven posterior inference. The resulting paper is centered on showing how a probabilistic surrogate can support an integrated uncertainty-to-risk and uncertainty-to-feasible-region workflow in a multi-physics engineering system.

本文建立了一套面向耦合 HPR 系统的不确定性-风险分析端到端流程。该流程采用概率神经代理模型逼近高保真的中子学-热学-力学映射，随后利用该代理模型完成三项相互关联的任务：在固定输入域下进行正向不确定性传播、开展应力风险分析以及执行观测驱动的后验推断。本文的核心重点在于展示概率代理模型如何支撑多物理工程系统中“从不确定性到风险、再到可行域”的一体化流程。

## 1.1 复杂耦合系统，不确定性参与决策；Complex coupled systems and uncertainty-aware decision making

uncertainty → affects safety → affects decision

## 1.2 高保真不确定性计算的瓶颈 Computational bottlenecks of high-fidelity uncertainty analysis

## 1.3 gap：传统代理模型

Existing surrogate models are primarily designed for prediction accuracy, rather than for uncertainty-to-risk mapping and decision support.

现有代理模型大多面向预测精度，而不是面向 uncertainty-to-risk mapping、safety-threshold interpretation 和 decision support。

## 1.4 研究现状：Representative reactor case and unresolved workflow gap

## 1.5 当前工作贡献和文章布局安排 This work and contributions

We establish an end-to-end computational workflow that maps input uncertainty to risk quantification and safety-constrained parameter regions.

# 2. Results[报结果，给数字，一句话解释不展开，This result motivates the subsequent analysis of ...]

## 2.1 Model performance and final model choice

**English**

We first assess whether the probabilistic surrogate is accurate enough to support downstream uncertainty-to-risk analysis. Physics -regularized added model provides the most balanced trade-off between predictive accuracy and probabilistic sharpness. Relative to the normal baseline BNN, Physics -regularized added model reduces the test negative log-likelihood from 0.4067 to 0.1785, lowers the mean RMSE over the primary outputs from 3.907 to 3.707, and improves the mean $R^2$from 0.759 to 0.775. For the two main observables, the stress RMSE decreases from 9.364 MPa to 8.600 MPa and the stress $R^2$increases from 0.913 to 0.927, while the keff RMSE decreases from $2.94 \times 10^{-4}$to $2.71 \times 10^{-4}$and the keff $R^2$increases from 0.829 to 0.855. At the same time, the mean 90% prediction interval width over the primary outputs narrows from 15.92 to 11.86, although the corresponding mean PICP90 decreases from 0.957 to 0.927. This pattern indicates that Physics -regularized added model is less conservative but more concentrated, and it is therefore adopted as the main constrained surrogate in the subsequent analyses.

**中文**

我们首先评估概率代理模型是否足以支撑后续的“不确定性—风险”分析链条。在主文比较的模型中，Level 2 在预测精度与概率区间收敛性之间表现出更均衡的折中。相较于 Level 0，Level 2 将测试集负对数似然从 0.4067 降至 0.1785，将主输出平均 RMSE 从 3.907 降至 3.707，并将平均 $R^2$从 0.759 提升至 0.775。对于两个核心观测量，最大应力的 RMSE 从 9.364 MPa 降至 8.600 MPa，$R^2$ 从 0.913 提升至 0.927；keff 的 RMSE 从 $2.94 \times 10^{-4}$ 降至 $2.71 \times 10^{-4}$，$R^2$ 从 0.829 提升至 0.855。与此同时，主输出平均 90% 预测区间宽度由 15.92 收窄至 11.86，但对应的平均 PICP90 也由 0.957 降至 0.927。这个结果表明，Level 2 的区间更集中、保守性略低，因此本文后续分析采用 Level 2 作为主约束代理模型，并以 Level 0 作为无约束基线。

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## 2.2 Forward uncertainty propagation and stress-risk quantification

**English**

To quantify how input uncertainty translates into safety-relevant outcomes, we propagated the eight uncertain empirical parameters through the trained surrogates using 20,000 Monte Carlo samples. For Level 0, the propagated distribution of iteration-2 maximum global stress has a mean of 154.04 MPa, a standard deviation of 47.16 MPa and a 95th percentile of 238.72 MPa. For Physics -regularized added model, the propagated distribution of steady state maximum global stress has a mean of 156.63 MPa, a standard deviation of 48.54 MPa and a 95th percentile of 245.01 MPa. The propagated keff distribution remains much tighter, with mean of 1.10382 and standard deviation of only $1.67 \times 10^{-3}$for Levels 0 and 2, respectively. The overall primary-output CVR is 0.657 for Level 0 and 0.667 for Level 2. Together, these results show that uncertainty is not distributed uniformly across outputs: the stress response exhibits broad dispersion at the system level, whereas keff remains comparatively concentrated. [要加入对比 iter1，iter2，也就是说要对比两次计算的均值变化，可以描述为系统性偏差，不确定性变化可以描述为放大缩小？；不要 level0，只要 level2，并且名字是 Physics -regularized added model，或者你有什么其他更合适的名字推荐吗，iteration-2 改名为 steady-state，对应的 iter1 大概是叫串行计算未反馈的计算结果，你想一想取一个什么名字好，全文统一，然后避免使用第一次串行计算和第二次耦合计算，就是一个是未反馈，第二个是耦合稳态，不要解释说是计算了两遍之类的，然后 keff 是一个很精细的量，需要精确到十的负五次方，也就是说$1.67 \times 10^{-3}$*应该表示为*$167pcm$，其实也是比较大了]

中文

为量化输入不确定性如何传递到安全相关输出，本文将 8 个不确定经验参数通过训练后的代理模型进行 20,000 次 Monte Carlo 正向传播。对于 Level 0，第二次迭代最大整体应力的传播分布均值为 154.04 MPa，标准差为 47.16 MPa，95% 分位数为 238.72 MPa；对于 Level 2，对应数值分别为 156.63 MPa、48.54 MPa 和 245.01 MPa。相比之下，keff 的传播分布明显更集中，其均值分别为 1.10379 与 1.10382，标准差仅为 $1.66 \times 10^{-3}$和 $1.67 \times 10^{-3}$。主输出整体 CVR 分别为 0.657（Level 0）与 0.667（Level 2）。这些结果共同表明，不确定性并不会在各输出之间均匀分配：应力响应在系统层面表现出明显更宽的分布，而 keff 则保持相对集中的波动范围。

English

Using the external stress thresholds of 110, 120 and 131 MPa, we further translate the propagated stress distributions into exceedance-based risk metrics. Under the main threshold of 131 MPa, the forward-UQ exceedance probability is 0.6396 for Level 0 and 0.6453 for Level 2. Under the stricter 120 MPa threshold, the corresponding risks increase to 0.7243 and 0.7326, and under 110 MPa they further increase to 0.8026 and 0.8118. Thus, the two models produce a qualitatively consistent safety picture, but the regularized Level 2 model yields a slightly more conservative high-stress tail. This result motivates the threshold-based interpretation developed in the next subsection.

中文

【这个三个阈值对比不要写在主论文里了，全部挪到附录，主论文就好像是我们只做了一次针对 ss316 失效的 131mpa 的分析一样的，你把全文需要改了的地方都标注出来】在 110、120 和 131 MPa 三个外加应力阈值下，我们进一步将应力传播分布转化为基于超限概率的风险量。以主阈值 131 MPa 为例，Level 0 与 Level 2 的 forward-UQ 超限概率分别为 0.6396 与 0.6453；当阈值收紧到 120 MPa 时，对应风险升至 0.7243 与 0.7326；进一步收紧到 110 MPa 时，风险增至 0.8026 与 0.8118。由此可见，两类模型给出的整体安全图景在定性上是一致的，但加入约束后的 Level 2 在高应力尾部给出了略偏保守的估计。这一结果为下一节的阈值依赖风险解释提供了直接动机。

English

The threshold dependence becomes even clearer on the held-out test set. For Level 2, the predictive exceedance probability under 131 MPa is 0.8274, compared with 0.8476 from the posterior mean alone and 0.8293 from the ground-truth stress values. When the threshold is reduced from 131 to 120 and 110 MPa, the predictive exceedance

probability rises from 0.8274 to 0.9036 and 0.9553, respectively. The corresponding mean predicted stress is 162.60 MPa, close to the true mean of 162.96 MPa, with an average predictive stress uncertainty of 8.68 MPa. These results show that risk is not equivalent to variance or mean shift alone; instead, it is a threshold-dependent transformation of the predictive distribution. 【The engineering basis of the 131 MPa threshold should still be stated formally in Methods or Supplementary Information.】

**中文**

在独立测试集上，这种阈值依赖性表现得更为明显。以 Level 2 为例，在 131 MPa 主阈值下，predictive 超限概率为 0.8274，而仅使用后验均值判定时为 0.8476，真实应力样本给出的超限比例为 0.8293。当阈值从 131 MPa 进一步收紧到 120 MPa 和 110 MPa 时，predictive 超限概率分别升至 0.9036 和 0.9553。对应的平均预测应力为 162.60 MPa，与真实均值 162.96 MPa 接近，平均预测不确定度为 8.68 MPa。这些结果表明，风险不能简单等同于方差大小或均值偏移本身，而应理解为预测分布在外加阈值作用下的转换结果。【131 MPa 的工程依据仍建议在 Methods 或 Supplementary 中单独正式说明。】

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## 2.3 Variance-based sensitivity attribution and uncertainty amplification

**English**

We next examine which uncertain inputs govern the stress and keff responses. For iteration-2 maximum global stress, the first-order Sobol index is dominated by $E_{\text{intercept}}$, with $S_1 = 0.709$for Level 0 and $S_1 = 0.626$for Level 2. The second most important contributor is $\alpha_{\text{base}}$, with $S_1 = 0.111$and 0.119, respectively. All remaining inputs contribute much less at first order, typically at or below the $10^{-2}$ – $10^{-1}$ level. The total-effect indices show the same ranking: for stress, $ST(E_{\text{intercept}}) = 0.739$at Level 0 and 0.669 at Level 2, while $ST(\alpha_{\text{base}}) = 0.126$and 0.143. The repeated-estimation confidence intervals remain narrow enough to preserve this ordering for the dominant terms, indicating that the main attribution result is statistically stable under the present Sobol setting.

**中文**

随后我们分析哪些不确定输入主导了应力与 keff 【第一句话不说这两个参数】的响应变化。对于【补充是对于安全很重要的量，】第二次迭代最大整体应力，一阶 Sobol 指数明显由 $E_{\text{intercept}}$主导：Level 0 下 $S_1 = 0.709$，Level 2 下 $S_1 = 0.626$。第二重要的贡献项为 $\alpha_{\text{base}}$，对应 $S_1$分别为 0.111 和 0.119。其余输入在一阶贡献上都明显更小，通常位于 $10^{-2}$–$10^{-1}$ 量级。总效应指数给出的排序也一致：对应力而言，$ST(E_{\text{intercept}})$ 在 Level 0 与 Level 2 下分别为 0.739 与 0.669，而 $ST(\alpha_{\text{base}})$分别为 0.126 与 0.143。在当前 Sobol 设定下，重复估计得到的置信区间足以保持主导项排序稳定，说明该主归因结果在统计上是稳健的。 【并且补充 iter1 iter2 的数值计算对比，蜻蜓点水讨论一句，注意写法和我上面的要求一样，例如不要 level0 2，并且改名字，罗列数字部分尽可能精简】

English

For iteration-2 keff, the dominant pathway is different. The largest first-order contribution comes from $\alpha_{\text{base}}$, with $S_1 = 0.746$for Level 0 and $S_1 = 0.717$for Level 2, followed by $\alpha_{\text{slope}}$, with $S_1 = 0.162$and 0.201. The total-effect indices again preserve this ordering, with $ST(\alpha_{\text{base}}) = 0.763$and 0.727 and $ST(\alpha_{\text{slope}}) = 0.163$and 0.218 for Levels 0 and 2. In contrast to the stress response, the stiffness-related parameter $E_{\text{intercept}}$contributes only weakly to keff. This separation between a stiffness-dominated stress pathway and a thermal-expansion-dominated keff pathway provides a quantitative basis for the mechanistic discussion developed later in the manuscript.

中文

【并且补充 iter1 iter2 的数值计算对比，蜻蜓点水讨论一句，注意写法和我上面的要求一样，例如不要 level0 2，并且改名字，罗列数字部分尽可能精简】对于第二次迭代 keff【补充是对于核安全物理控制相关很重要的量】，主导路径则明显不同。最大的一阶贡献来自 $\alpha_{\text{base}}$，其 $S_1$在 Level 0 与 Level 2 下分别为 0.746 和 0.717；其次为 $\alpha_{\text{slope}}$，对应 $S_1$分别为 0.162 和 0.201。总效应指数同样保持这一排序：$\alpha_{\text{base}}$ 的 $ST$分别为 0.763 与 0.727，$\alpha_{\text{slope}}$ 的 $ST$分别为 0.163 与 0.218。与应力响应不同，刚度相关参数 $E_{\text{intercept}}$对 keff 的贡献仅为弱影响。也就是说，应力主导于“刚度路径”，而 keff 更受“热膨胀路径”控制，这为后文的机理讨论提供了明确的定量基础。

【results 这里只用一句，说明几何膨胀反馈影响比较剧烈？大概这样？你有更好的想法吗】

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【这 2.3 2.4 中间还差了一节，是类似于李冬那样的，参数修正结果检验，想一下怎么和主线融合在一起，就是她的论文中：实验-我的高保真计算，她的代理模型程序对应我的 BNN 代理模型，然后她根据实验修正程序输入参数的参数，我根据高保真计算修正 BNN 输入参数；我还要补实验，并且逻辑连贯的把这一段放进去，例如还是能够和 risk，应力，keff，不确定性完整逻辑链条，最好是多一些。哦哦，大概是和 2.4 的这一部分独立出来，不要两个混在一起，当然也可能是因为我没放数据的原因，ok，那还有就是要在补充实验的时候记录时间，或者还有什么其他表示复杂度的参数或者图片，证明简化是有必要的并且有效果的】

## 2.4 Observation-driven posterior inference and safety-feasible region analysis

本文当前阶段更适合把 inverse 模块定位为：

（1）在 synthetic-observation 条件下验证后验推断与参数收缩能力；

（2）进一步探索其作为工程经验参数精细化工具的潜力，即将后验代表点回代高保真模型，检验更新参数是否更符合目标响应；

（3）在更进一步的层面上，利用安全约束下的可行参数区域，为后续材料筛选与目标性能设定提供定量依据。

English

We finally examine whether observation-driven posterior inference can translate stress constraints into parameter-space restriction. The reduced inverse benchmark is carried out on 20 held-out synthetic cases using four calibration parameters. Across these cases, the mean acceptance rate is 0.628, the mean absolute observation-fit error is 1.227, and the mean observable coverage90 is 1.0. The mean posterior safety-feasible fraction is 0.1667 under the 131 MPa threshold, 0.0792 under 120 MPa, and 0.0373 under 110 MPa. These averages

already indicate that tighter safety criteria strongly compress the admissible posterior region. 【At present, the uploaded comparison table contains only the reduced run; if a full inverse benchmark is added later, this paragraph should be expanded into an explicit full-versus-reduced comparison.】

中文

最后我们考察：基于观测的后验推断能否将应力约束转化为参数空间中的可行域收缩。当前 reduced inverse benchmark 在 20 个独立 hold-out synthetic case 上进行，并仅反演 4 个校准参数【要说清楚哪 4 个】。在这些 case 上，平均接受率为 0.628，平均观测拟合绝对误差为 1.227，平均 observable coverage90 为 1.0。在 131 MPa 阈值下，后验安全可行域的平均比例为 0.1667；当阈值收紧到 120 MPa 与 110 MPa 时，这一比例进一步降至 0.0792 与 0.0373。这些平均结果已经表明，越严格的安全要求会显著压缩后验中的可接受参数区域。【目前上传的比较表只包含 reduced 结果；若后续补充 full inverse benchmark，这一段应升级为 full 与 reduced 的正式对比。嗯嗯要补充的】 【这个三个阈值对比不要写在主论文里了，全部挪到附录，主论文就好像是我们只做了一次针对 ss316 失效的 131mpa 的分析一样的，你把全文需要改了的地方都标注出来】

English

The case-wise results show the same trend more clearly. When the observed stress is 96.34 MPa, the feasible posterior fraction at 131 MPa is still 1.0. As the observed stress increases to 119.44, 127.34 and 133.39 MPa, this fraction drops to 0.8725, 0.6692 and 0.5508. Once the observed stress enters the clearly unsafe regime, the feasible fraction collapses rapidly: it is 0.0167 at 145.35 MPa, 0.0058 at 147.9 MPa, and becomes zero for several cases above 155 MPa, including 155.34, 158.66, 165.8 and 244.17 MPa. Thus, the posterior does not merely shift continuously with the observation; instead, the safety-feasible region contracts sharply once the observed stress crosses the neighborhood of the main threshold.

中文

分 case 的结果更清楚地展示了这一趋势。当观测应力为 96.34 MPa 时，在 131 MPa 阈值下，后验可行比例仍为 1.0；当观测应力增加到 119.44、127.34 和 133.39 MPa 时，这一比例分别降至 0.8725、0.6692 和 0.5508。一旦观测应力进入明显不安全区间，可行比例就会迅速塌缩：在 145.35 MPa 时仅为 0.0167，在 147.9 MPa 时为 0.0058，而对于多个高于 155 MPa 的 case（如 155.34、158.66、165.8 和 244.17 MPa），可行比例已降为 0。也就是说，后验并不是随着观测平滑平移，而是在观测应力接近并越过主阈值附近后，安全

可行域发生了明显收缩。 【我没看懂这个可行比例是什么？你先把我解释通，然后再想想怎么把这部分放进论文里，应该不叫这个名字吧？】

English

Posterior predictive agreement remains strong at the observable level. The mean absolute posterior predictive error is $2.58 \times 10^{-4}$for iteration-2 keff, 1.84 for iteration-2 maximum fuel temperature, 2.10 for iteration-2 maximum monolith temperature, 2.19 MPa for iteration-2 maximum global stress, and $8.28 \times 10^{-4}$for iteration-2 wall2. All five observables achieve coverage90 of 1.0. At the parameter level, however, recoverability is heterogeneous: $\alpha_{\mathrm{base}}$and $\alpha_{\mathrm{slope}}$both achieve coverage90 of 0.85 with relatively narrow 90% widths, whereas $SS316_k_{\mathrm{ref}}$reaches coverage90 of only 0.7, and $E_{\mathrm{intercept}}$remains much broader despite coverage90 of 1.0. Therefore, the inverse analysis is better interpreted as a posterior restriction and feasible-region analysis than as uniformly precise parameter recovery for all calibrated inputs.

中文

在观测量层面，后验预测拟合仍然保持较强一致性。其平均后验预测绝对误差分别为：【注意改名不要 iter2 哈】iteration-2 keff 【注意换乘 pcm 哈】为 $2.58 \times 10^{-4}$，iteration-2 最大燃料温度为 1.84，iteration-2 最大整体应力为 2.19 MPa，iteration-2 最大 monolith 温度为 2.10，以及 iteration-2 wall2 为 $8.28 \times 10^{-4}$。这五个观测量的 coverage90 均为 1.0【这个效果好像不好？那么不好的话可不可以不说？或者我怎么改进？还是说我没懂是什么的观测值？解释一下把我说懂先】。然而在参数层面，恢复性呈现明显异质性：$\alpha_{\mathrm{base}}$ 和 $\alpha_{\mathrm{slope}}$的 coverage90 均为 0.85，且 90% 区间宽度相对较窄；$SS316_k_{\mathrm{ref}}$ 的 coverage90 仅为 0.7；而 $E_{\mathrm{intercept}}$虽然 coverage90 达到 1.0，但后验区间仍然很宽。因此，这里的 inverse 分析更适合被解释为“后验约束与可行域收缩分析”，而不是“所有校准参数都能被同等精确地恢复”。 【这里看起来很绕。并且没有给出一个比较清晰的结论，例如是不是与敏感性分析的结果相挂钩相关联之类的】

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## 2.5 Posterior-informed parameter refinement and safety-feasible region screening

A：后验更新可用于参数精细化
B：进一步可用于安全约束下的参数要求反推

## 2.6 【Optional】Computational implication of the surrogate workflow

**English**

A dedicated wall-clock benchmark was also carried out to assess the practical computational role of the surrogate workflow. Under the standard execution environments used in this study, one representative high-fidelity coupled run required about 2268 s in the recorded benchmark. This confirms that direct many-query uncertainty propagation or repeated inverse benchmarking with the original coupled solver would be prohibitively expensive in practice. 【The final manuscript should insert the measured surrogate single-query latency, campaign-level wall-clock estimates, and the corresponding speedup factors once the benchmark table is frozen.】【代理模型计算的我应该也上传了。还有我还想比较简化的 mcmc 反向优化和全参数的优化时间，适合放在这里吗？】

**中文**

本文还进行了专门的 wall-clock 基准测试，以评估代理工作流在实际计算中的作用。在本研究所采用的标准执行环境下，一次代表性的高保真耦合计算在记录基准中约需 2268 s。这个结果说明，如果直接使用原始耦合求解器去完成 many-query 正向不确定性传播或 repeated inverse benchmark，在实际计算上将非常昂贵。【正式稿中仍建议在速度基准表最终冻结后，补入代理单次推理延迟、整体 campaign 耗时估计以及对应的 speedup 倍数。】

## 2.1 数据集与模型选择表现 Dataset and model selection

Probabilistic surrogates enable many-query analysis required for uncertainty-to-decision workflows.

- model performance → surrogate can be used for many-query analyses

## 2.2 前向不确定性传递和应力超限风险 Forward uncertainty propagation and stress-risk quantification

To quantify how input uncertainty translates into safety-relevant outcomes, we first perform forward uncertainty propagation...

- forward UQ → motivates threshold risk interpretation

## 2.3 Sobol 敏感性分析与不确定性传递分析 Variance-based sensitivity attribution and uncertainty amplification

- Sobol → motivates mechanistic discussion

## 2.4 基于观测的后验推断与安全可行域分析 Observation-driven posterior inference and safety-feasible region analysis

- inverse posterior → motivates feasible-region restriction discussion

# 3. Discussion[为什么，怎么样，说明什么]

## 3.1 安全相关可观测值的不确定性变化 Uncertainty amplification in safety-relevant observables

**Uncertainty is selectively amplified in safety-relevant observables, rather than uniformly propagated across outputs.**

### 3.1.1 应力

Uncertainty is not uniformly propagated across outputs; instead, stress emerges as an uncertainty-amplified observable under multi-physics coupling.

### 3.1.2 keff

- indicator of the feedback-adjusted coupled state。核控制安全相关重要参数

## 3.2 风险作为不确定性的阈值依赖转换 Risk as a threshold-dependent transformation of uncertainty

Uncertainty does not directly translate into risk; risk emerges as a threshold-dependent transformation of uncertainty.

### 3.2.1 a threshold-dependent transformation of it（？放在这里还是附录

## 3.3 不确定性传播的机理途径 Mechanistic pathways of uncertainty propagation

### 3.3.1 Sobol

### 3.3.2 几何/应力分析

### 3.3.3 geometry / thermal expansion / stiffness / conductivity

## 3.4 从后验推理到可行域约束 From posterior inference to feasible-region restriction

### 3.4.1 restrictions

Safety constraints impose restrictions on the parameter space.

- posterior contraction
- feasible region shrinkage
- safety constraints induce geometric restriction in parameter space

本文中的逆向推断并不直接等同于真实实验校准，而是首先在合成观测条件下验证后验推断的统计有效性。在此基础上，后验代表点可进一步作为参数更新候选，回代至高保真模型中检验其是否使目标响应更加一致。因而，该框架不仅可用于参数后验收缩，也可为工程经验参数的精细化修正提供方法依据。进一步地，在安全阈值约束下得到的可行参数区域，还可被解释为对材料相关参数的性能要求，为后续材料筛选或目标性能设定提供定量参考。

## 3.5 Scope, limitations and outlook

### 3.5.1 方法边界：不是 PINN，不是 first-principles closure，而是 structured probabilistic response model

### 3.5.2 局限：synthetic inverse、OOD 不强、prior 假设、131 MPa 仍需更强工程依据

### 3.5.3 展望：真实实验校准、prior sensitivity、mesh / HF refinement、prior–posterior 收缩分析

# 4. Methods

## 4.1 高保真核热力耦合计算 High-fidelity neutron–thermal–mechanical workflow

## 4.2 不确定性热工程序经验参数选择和输出标量设置 Uncertain empirical parameters and output definition

## 4.3 概率神经网络 BNN 与物理正则化设置 Probabilistic neural response model and physics regularization

### 4.3.1 likelihood（Gaussian NLL）

### 4.3.2 physics constraint（gradient）

## 4.4 正向不确定性量化传递与风险阈值设置 Forward UQ and threshold risk protocol

### 4.4.1 threshold 是 external constraint

### 4.4.2 risk = exceedance probability

## 4.5 基于方差分解的 Sobol 敏感性分析 Variance-based sensitivity analysis

## 4.6 基于高保真程序计算观测修正后验推断与可行域分析 Observation-driven posterior inference and feasible-region analysis

## 4.7 Frozen surrogate and benchmark separation

- surrogate training uses a frozen train/val/test split
- calibration pool is held out from surrogate preparation
- all downstream analyses use the same pretrained surrogate

- Figure 1: overall workflow with frozen surrogate + forward/inverse branches
- Figure 2: forward UQ + threshold risk
- Figure 3: iter1 vs iter2 + Sobol / uncertainty amplification
- Figure 4: full vs reduced inverse + feasible region
- Figure 5（可选）: posterior-informed parameter refinement / 2D feasible region

# Supplementary Information

S1. Additional model comparisons (including Level1/4)

S2. Multi-threshold robustness (110/120 MPa)

S3. OOD split and results

S4. Hyperparameter table

S5. MCMC diagnostics / trace / prior–posterior comparison【先验后验 感觉重要的？要不要放进 discussion】

S6. Additional per-output metrics